Retrograde Vacuum Friction: Front Compression and Trailing Suction in Hawking Evaporation Backgrounds

A phenomenological extension proposal for active tensor foam

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Purpose	Formulate a testable, continuous-density extension; supply an implementation sketch and verification plan.
Reference system	QVEX / NGC 6503 code and notes supplied by the user; tensor-model theory paper supplied as contextual background.
Status	Hypothesis and solver design only. Not an established Hawking-induced galactic force or a validated observational inference.

Executive summary

This note extends the project's radial foam description by separating (i) a leading-side compressive stress and (ii) a trailing-side tensile / suction stress sourced by a smooth, time-dependent evaporation-associated density field. The net radial force is computed by integrating the stress over a finite, oriented interaction kernel, rather than declaring that a local density alone is a force. A sign reversal can occur when the trailing contribution overtakes the front contribution. The supplied project's raw empirical friction expression crosses zero at approximately 12.10 kpc; that is a mathematical sign change in that expression, not evidence that Hawking evaporation creates the proposed stresses or that the physical force reverses at that radius.

Critical interpretation

The QVEX source explicitly keeps `foam_friction_raw` for regression comparison and identifies a non-negative exchange-current law as the production drag. Do not silently reinterpret the raw negative branch as a measured retrograde Hawking force. The extension below adds a separately named hypothesis channel and leaves the existing causal exchange current unchanged.

1. Scope and physical bookkeeping

The tensor-model reference (Halmagyi & Mondal, JHEP 07 (2018) 095) studies large-N tensor quantum mechanics and probe fields in a black-hole background, including propagators and chaotic correlators. It does not derive a galactic-scale Hawking evaporation density profile, front compression, trailing suction, or a 12.11 kpc transition. The connection here is therefore a proposed effective-medium analogy, not a consequence of that paper.

Hawking evaporation is a black-hole mass/energy-loss process. A spatially continuous radial profile around a galaxy is not a standard prediction: it must be specified as a source/transport model, with boundary conditions, conservation laws and a mechanism coupling it to matter. Define an effective perturbation `delta_rho_H(r,t)` as a phenomenological field, separate from the background vacuum density. Avoid treating the black-hole Hawking temperature as a halo-wide thermal bath.

2. Continuous profile and stress model

Start with the project's baseline profile and add a smooth evaporation-linked component:

```
rho_foam(r,t) = rho_popcorn(r) + delta_rho_H(r,t)
rho_popcorn = rho0 [1 + r/lambda_coh]^(-gamma_vac) exp(-r/R_cutoff)
delta_rho_H(r,t) = A_H(t) exp(-r/L_H) S(r; r_in, w_in) (illustrative; calibrate A_H, L_H,
r_in, w_in)
```

Here `A_H(t)` is a signed amplitude with explicit units of density, and may be connected to an evaporation luminosity through a separately justified coupling and transport equation. A differentiable inner window S can prevent a spurious central cusp. A minimal time law could use `dM_BH/dt = -L_H/c^2` and `A_H(t)=chi L_H(t)/c^3 L_*^(-2)` (one possible dimensional convention only); `chi` and transport are unknown and must not be implied by the input dataset.

Let a scalar effective stress be `p_eff(r,t)=p0 + c_eff^2 delta_rho_H(r,t)` or a constitutive law measured/calibrated independently. For a leading/trailing probe footprint, define a normalized, oriented kernel `K(x;ell_f,ell_t)`, with front lobe positive and trailing lobe negative. The radial force is the stress-work / pressure-gradient integral:

```
F_H(r,t) = - A_eff integral K(x) d[p_eff(r+x,t)]/dx dx
K = K_front - eta K_trail ; integral K_front dx = integral K_trail dx = 1
```

Alternatively for an extended surface, integrate the traction `-p_eff n_hat` over its boundary; do not add "front pressure" and "rear suction" as unrelated scalars without a consistent normal and sign convention. A local compression maps to a positive inward/outward radial component only after the probe orientation is specified. The kernel form makes finite spatial extent and smooth profiles explicit.

Define a chosen outward-positive radial convention. Separate the hypothesis channels as `F_net = F_exchange + F_front + F_trail`, where the legacy exchange force retains its sign and energy constraints. Report components and units independently. A retrograde transition radius is a root `r_*`: `F_net(r_*,t)=0` with an actual sign flip; a root of the old dimensionless radial formula is not automatically that root.

3. Proposed ActiveTensorFoamSolver extension

The short Python sketch below retains the project's density profile and existing exchange API; it adds a smooth evaporation perturbation, constitutive pressure, oriented finite-difference kernel, and an explicitly hypothesis-only net force. Inputs are in consistent but user-supplied units: density in g/cm^3, radius in kpc, pressure/force scaling in the solver's chosen system. The coupling prefactor must carry units that convert the integral to force. Numerical values shown are test fixtures, not measured Hawking parameters.

```
import numpy as np
from dataclasses import dataclass

@dataclass
class RetrogradeFoamExtension:
    base: object                # ActiveTensorFoamSolver
    A_H: float = 0.0            # g/cm^3; set from an external model
    L_H: float = 8.0            # kpc; illustrative scale, not fitted
    r_in: float = 0.2           # kpc; smooth inner-window midpoint
    w_in: float = 0.1           # kpc; positive width
    c_eff2: float = 1.0         # pressure / density in a consistent system
    eta_trail: float = 1.0      # dimensionless trailing-lobe weight
    ell_front: float = 0.5      # kpc; kernel width
    ell_trail: float = 1.0      # kpc; kernel width
    force_scale: float = 1.0    # units conversion; calibrate, never assume

    def rho_h(self, r, t=0.0):
        r = np.asarray(r, float)
        if np.any(r < 0) or np.any(~np.isfinite(r)):
            raise ValueError("r must be finite and non-negative")
        window = 1.0 / (1.0 + np.exp(-(r-self.r_in)/self.w_in))
        return self.A_H * np.exp(-r/self.L_H) * window

    def p_eff(self, r, t=0.0):
        return self.c_eff2 * self.rho_h(r, t)

    def front_trail_force(self, r, t=0.0, dx=0.01):
        # Radial pressure-gradient response; kernel balance avoids arbitrary offset.
        x = np.arange(-5*max(self.ell_front,self.ell_trail),
                      5*max(self.ell_front,self.ell_trail)+dx, dx)
        def gauss(x, ell):
            return np.exp(-0.5*(x/ell)**2)/(np.sqrt(2*np.pi)*ell)
        K = gauss(x, self.ell_front) - self.eta_trail*gauss(x, self.ell_trail)
        gradp = (self.p_eff(np.maximum(r+x+dx,0),t) -
                 self.p_eff(np.maximum(r+x-dx,0),t))/(2*dx)
        return -self.force_scale*np.trapz(K*gradp, x)

    def net_hypothesis_force(self, r, dv=1.0, t=0.0):
        legacy = self.base.exchange_friction(r, dv)  # unchanged, dissipative
        return legacy + self.front_trail_force(r, t)
```

Implementation cautions

In production code, vectorize the kernel integration, enforce a boundary treatment that does not truncate kernel normalization at r=0, expose the time evolution `A_H(t)`, and document the force units. Use a finite-volume stress balance when possible. Do not infer `A_H` from the 12.11 kpc zero crossing. Include the leading/trailing components in solver output and keep `net_hypothesis_force` outside any validated production path until the tests below pass.

A genuine `ActiveTensorFoamSolver` extension can use inheritance or composition; the sketch uses composition so the existing tested solver remains untouched. The project's negative-radius validation helper should also be reused for every radial input, and numerical differentiation needs convergence tests at several grid spacings.

4. Verification plan and parameters

First verify the legacy reference expression separately from the new physics. The project defaults are `rho0=7.02e-30 g/cm^3, lambda_coh=12 kpc, gamma_vac=1, R_cutoff=50 kpc`, and friction parameters `scale=1.1, r_core=4.5 kpc, penalty=0.12`. Numerically, this documented raw expression has root 12.1005 kpc. Values may differ from a rounded 12.11 kpc depending on rounding/solver and source revisions.

Parameter / check	Verification setting or acceptance criterion
Legacy zero-crossing	bracket [10, 15] kpc; root ≈ 12.1005 kpc; verify raw(r<root)>0 and raw(r>root)<0 nearby.
Continuous density	rho_popcorn(0)=rho0; rho_H smooth, finite, decays at large r; sensitivity scan A_H=0 and signed small perturbations.
Kernel geometry	ell_front=0.5 kpc, ell_trail=1.0 kpc, eta=1 as unit-test fixtures; check normalized lobes and orientation reversal.
Resolution	dx=0.02, 0.01, 0.005 kpc; force/root converge within an explicitly declared tolerance (e.g. 1%); halve domain-step error.
Conservation / energy	front + trail reaction forces balance in an isolated pair; legacy exchange power remains -Gamma dv^2 ≤ 0. Any energy source is accounted to BH/field.
Retrograde root	Find sign-changing roots of F_net on an announced interval. Claim transition near 12.11 kpc only if the new total force root converges there across resolution/parameter uncertainty.
Null model	Set A_H=0; new evaporation contribution must be identically zero. Compare to baseline without regression changes.

Useful automated assertions

```
assert np.isclose(base.vacuum_density(0), base.rho_0)
assert np.all(base.exchange_coefficient(np.linspace(0,50,101)) >= 0)
assert np.allclose(ext.rho_h(np.linspace(0,50,101)), 0.0)  # when A_H=0
assert np.isfinite(ext.front_trail_force(12.0, dx=0.01))
# Require convergence and a genuine sign change before reporting a new r_star.
# Assert momentum-pair balance and account for all field/BH energy exchange.
```

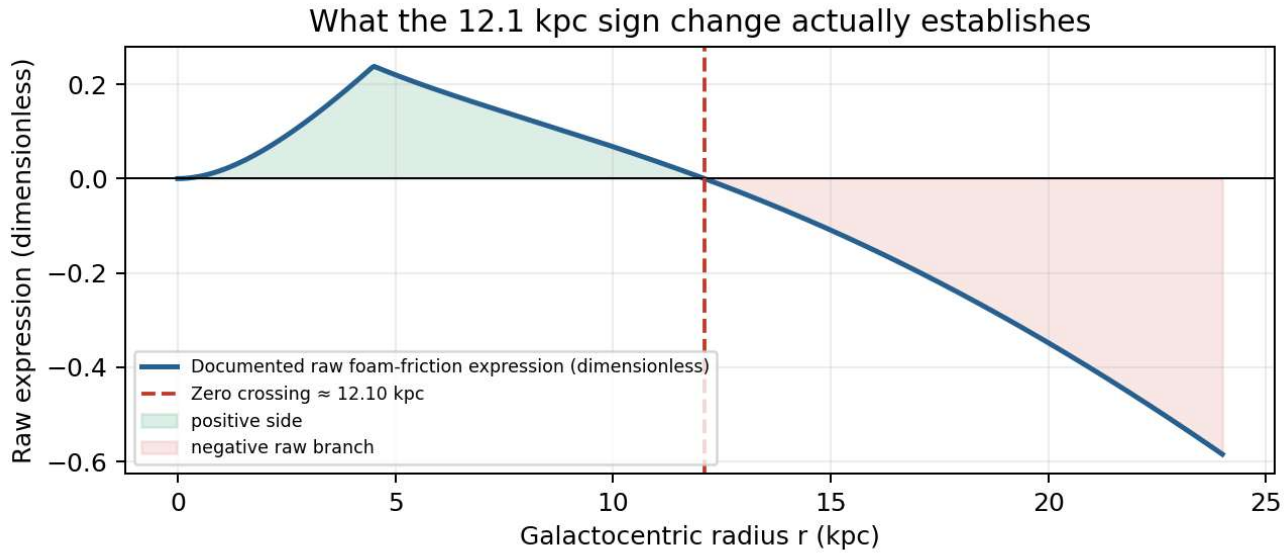


Figure 1. The curve is the project's raw friction expression only. The red dashed line marks its computed zero (~12.10 kpc). Shaded negative values diagnose its anti-damping sign, not a validated physical retrograde force.

5. Interpretation, falsifiability, and limitations

Leading-front compression: model this as a resolved, oriented positive pressure lobe. A compression is not inherently an inward force; the normal direction and radial projection determine the sign. Measure or derive the effective stress from a microscopic model, rather than assigning a sign by verbal analogy.

Trailing suction: represent as a tensile/stress deficit behind the moving probe, with equal-and-opposite reaction and an energy/momentum budget. “Suction” is not a separate attractive force unless a constitutive stress tensor and coupling to matter establish it.

Hawking profile: spatial smoothness is a modeling choice, not derivable from the evaporation statement alone. Specify BH mass, luminosity history, transport law, boundary conditions, coupling, dimensions, and uncertainty. For astrophysical black holes, evaporation is extremely slow; any claim of a significant galactic-scale density/force effect needs a quantitative magnitude estimate and comparison to other sources.

12.11 kpc: source notes state the raw expression changes sign near this scale and explicitly warn it is not the production drag law. Our independent numerical root is 12.1005 kpc using listed nominal coefficients. A retrograde net-force transition beyond 12.11 kpc would require the new stress term to overwhelm the dissipative exchange term in a calibrated total force and survive parameter, data, unit, and resolution tests. The current uploaded metrics are not a validated QVEX velocity fit.

Falsifiable outputs required before physical interpretation

1. A derivation or independent calibration for `A_H(t)`, the stress constitutive relation, and kernel scales.
2. Unit-consistent force, energy and momentum accounting, including black-hole/field reaction terms.
3. Convergence and null tests, and uncertainty bands on `r_*`.
4. Comparison against a baseline using held-out observables, with parameter degeneracies and model selection reported.
5. A solver implementation that does not alter the legacy causal exchange-force semantics.

References and provenance

Halmagyi, N. & Mondal, S. (2018), “Tensor models for black hole probes,” JHEP 07 (2018) 095, arXiv:1711.04385. Supplied PDF. Its subject is large-N tensor quantum mechanics and black-hole probe correlators; no Hawking radial density law is asserted here.

QVEX-NGC6503 project inputs supplied with this task: `nodes/active_tensor_foam.py`; `nodes_analysis/01_Vacuum_Popcorn_Density_Node.md`; `nodes_analysis/02_Nonlinear_Foam_Friction_Node.md`; `NGC6503_QVEX_Complete_Analysis.md`; `ngc6503_metrics.json`. Project source labels the friction sign change as a raw diagnostic, retains the non-negative exchange current as production law, and states that available baryonic residual metrics are not a QVEX prediction/fit.

Calculation check: Brent root of $1.1 \ln(1+r/4.5)[1-\exp(-r/12)] - 0.12 \max(0,r-4.5) = 0$ on [10,15] kpc. This reports only the nominal analytic expression’s sign crossing.